

meshClaw-KDP-KU Agent - Business Overview

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Status: Active (production, in daily use)

Interface: Slack (meshClaw-KDP-KU) | This doc updated: August 28, 2026

1. Background & Problem

The Athena Analytics team answers data questions about KDP (Kindle Direct Publishing) and Kindle Unlimited (KU), covering royalties, reading behavior, catalog health, fraud, and program performance. Historically, answers depended on scattered notebooks, one-off SQL files, and individual analysts' knowledge, making them slow, hard to reproduce, and person-dependent.

The problem is compounded by KDP/KU's data complexity: 30+ tables with non-obvious join keys, channel-specific payment logic, and silent pitfalls (e.g., weekly-vs-monthly royalty row fan-out over-counts ~4-5×, `ku_royalty_usd` doesn't populate until ~45 days post-month-end, "Active" subscriptions that are actually paused). Recurring deliverables like the monthly Fund Doc required QuickSight dashboards, Kerberos tickets, mandated SQL, and hand-transcription into Excel, making them slow and error-prone.

The meshClaw-KDP-KU agent was built to fix both problems. It runs on MeshClaw (ASR-approved with Claude Sonnet 4-6 model powering it) on the team's Cloud Desktop, 24/7 via Slack. Its knowledge base is AthenaAnalysisForge, a markdown repo of prior analyses and documented conventions. When the agent answers a question, it checks that repo first and writes new findings back, so the knowledge base grows with use.

2. Scope

In scope:

- Borrows and Qualified Borrows, KENP reads, Per-Page Rate (PPR), royalties, All-Star bonus
- KU subscriptions, churn, and membership lifecycle
- Catalog health: Select enrollment, live titles, business-model transitions (KDP ↔ Managed ↔ APub)
- Sales rank / Zeitgeist, glance views, and discovery

- Device mix and reading behavior
- Minerva fraud invalidations (Session Invalidator and End-of-Month)
- Publisher / vendor metrics and cross-format analysis (ebook, print, audiobook/VVAB)

Out of scope: Retail, advertising, Alexa, and other non-KDP/KU domains.

Operating constraints: read-only (SELECT only), documented tables only, US marketplace by default, default LIMIT on exploratory queries.

3. How It Works

3.1

The agent checks the analysis catalog for an existing answer before writing any SQL. If it needs a new query, it pulls context from `table_reference.md` and `data_caveats.md`, drafts SQL, and confirms with the user before running anything against Redshift. Results get written back to the repo as structured analyses, so repeated questions are faster.

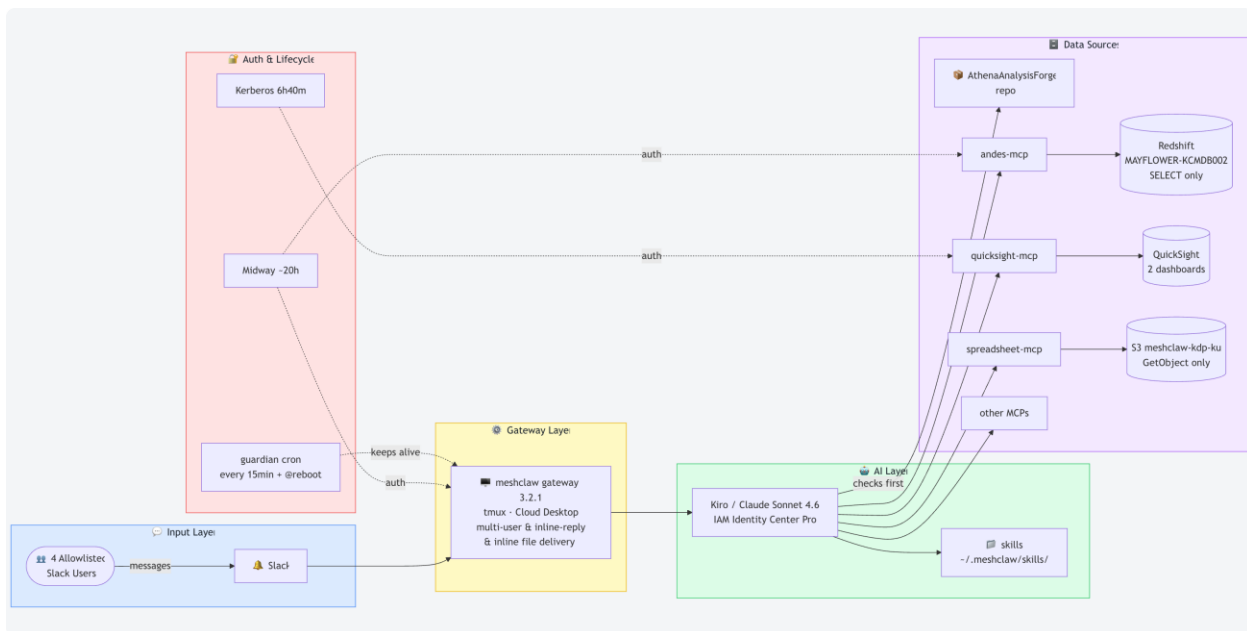


Illustration 1: Agent Architecture

3.2:

The biggest win so far: The agent calculates all five sections of the monthly KDP Select Fund Doc end to end, including KDPS Health Metrics (13 marketplaces) and the Tandem Fund Recommendation, and writes the results directly into the Fund Doc Excel workbook

via spreadsheet-mcp. That workflow used to take hours of manual SQL, dashboard lookups, and copy-paste into Excel.

Tool	Capability
Andes / Data Central Workbench	Read-only SQL on Redshift
quicksight-mcp	QuickSight dashboard data
spreadsheet-mcp (custom)	Read/write Excel workbooks
cradle-mcp, datanet-mcp	Pipeline monitoring / ETL
bumblebee-mcp	Metadata, lineage, compliance
midas-stargate-mcp, books-ai-mcp	Books/Stargate data services
builder-mcp	Code review, search, tickets
ai-community-slack-mcp, playwright-mcp	Slack integration; browser automation

Current deliverables:

- Monthly KDP Select Fund Doc (all five sections, written into the Excel workbook)
- KDP Select Penetration Dashboard (five monthly metrics: enrollment, retention, borrow share, KENP share, glance-view share)
- ~17 cataloged analyses across four domains (kdp_ku_growth, rosetta, KUA_Tandem, books_level_analysis)

Safety: all queries pass through a read-only wrapper (redshift.py); no DDL or writes to the warehouse; every action is audit-logged.

4. Upcoming Work

- Broaden spreadsheet-MCP automation to more recurring Excel deliverables beyond the Fund Doc
- Automate Kerberos renewal (most common daily breakage for QuickSight-backed metrics)
- Auto-sync new data caveats to the broader Athena/Stargate repo
- Consolidate schema sources (local cache, Minerva, Bumblebee) into a single source of truth
- Untrack binary/CSV deliverables under analyses/ (Git LFS migration, needs team coordination)

Analyses in flight: KU fund fairness, KU audio-launch impact, e-reader deprecation impact, KDP↔Managed transition performance, Stargate master royalties, sponsored-ads CTR drop, per-reader KENP decline.

5. Appendix

- Cloud Desktop: dev-dsk-pgtm-2b-7386cfa2.us-west-2.amazon.com (gateway in tmux, pinned MeshClaw version)
- Warehouse: Redshift MAYFLOWER-KCMDB002, role amzn/sdo/@pgtm, via Data Central Workbench / Andes
- Access: four authorized Slack users (owner + three teammates); model backend via AWS IAM Identity Center
- Key schemas: kdp_ddl, kcm_ddl, kcbi_ddl, crm_bi_ddl, subi_ddl, andes.kindle_cm_ddl

Reference links:

- AthenaAnalysisForge repo: code.amazon.com/packages/AthenaAnalysisForge
- meshClaw wiki:
w.amazon.com/bin/view/FinanceAutomation/GREFTech/AICoding/tools/meshclaw
- SOP for the meshClaw-KDP-KU agent: [meshClaw_Slack_Agent_SOP.docx](#)